

ADS1299 EEG Headset Design Notes

Component List

- ADS1299 IC (primary AFE)
- Teensy 4.1 microcontroller
- TPS7A02 LDO regulators (2.5V analog, 1.8V digital)
- TPD1E05U06 ESD protection diodes (20 units)

Analog Front-End Design

Input Protection and Filtering

- Differential input filtering using 22k Ω resistors and 0.22 μ F capacitors per channel for anti-aliasing and noise reduction
- ESD protection via TPD1E05U06DYAR diodes (0402 package) on each input line
- Mid-rail biasing at 1.25V using low-noise LDO with 10M Ω resistors to each input
- 100k Ω resistor from 1.25V reference to ground for bias current return path
- Guard rings around high-impedance input traces in PCB layout, connected to 1.25V reference to minimize leakage

Power Supply Architecture

Regulation and Decoupling

- Separate low-noise LDOs for analog (2.5V) and digital (1.8V) supplies
- 0.1 μ F ceramic capacitors close to each power pin
- 10-47 μ F bulk capacitors near LDO inputs for inrush current handling
- Star ground topology with single-point connection between analog and digital grounds
- Ferrite beads (600 Ω @ 100MHz) between power domains for high-frequency noise suppression

ADS1299 Configuration

Key Settings

- Internal 4.5V reference voltage (unless external reference needed)
- BIASOUT pin configured with 1M Ω current-limiting resistor
- AC lead-off detection circuits (if required)

- Input impedance: $1\text{G}\Omega$ in parallel with a few pF

Microcontroller Interface

Signal Integrity

- Logic level translation between 3.3V microcontroller and 1.8V ADS1299 using TXS0108E or similar
- $10\text{k}\Omega$ pull-up resistors on SPI lines
- Proper handling of START, RESET, and PWDN control lines per datasheet specifications

PCB Layout Strategy

Critical Considerations

- Analog components separated from digital circuitry
- Short, length-matched differential input traces
- Space reserved for optional EMI shielding cans
- Test points on power rails, SPI lines, and reference voltages for debugging

Testing and Validation

Verification Plan

- Time-domain simulations (200ms windows) for signal behavior
- FFT analysis for noise performance assessment
- Electrode impedance measurement using ADS1299 internal current sources
- Safety compliance verification for human use applications

References

Design based on best practices from OpenBCI, TI ADS1299 evaluation modules, and community documentation from openbci.com and [EEVblog](http://EEVblog.com) discussions.